AMINOACID CATABOLISM

INTRODUCTION:

- In the adult human body, only 20% of amino acids formed from protein degradation are catabolised to generate energy. The remaining 80% amino acids are used for protein biosynthesis.
- Further amino acids or proteins consumed excess are also used for energy production.
- Since most of amino acids formed from protein breakdown are recycled, to maintain normal body functions the diet should contain at least (only) amino acids or proteins (20%) that are used for energy production.
- In case, if excess is consumed that is also used for energy production because amino acids cannot be stored in the body.
- Under normal conditions, amino acids supply about 5-10% of total body energy.
- Amino acid catabolism is increased in starvation, diabetes and high protein diet.
- Intracellular synthesis of some amino acids also occurs. Depending on the cell needs they are utilized.
- In addition, amino acids are used for the formation of creatine, hormones, glutathione, purines, pyrimidines etc.

Amino acid catabolism occurs in two main stages.

1. First stage is the removal of amino group of amino acids as ammonia. Ammonia is converted to urea and excreted in urine.
2. In the second stage carbon skeletons of amino acids are converted into intermediates of TCA cycle and acetyl-CoA. Then they are either oxidized in TCA cycle for generation of energy or used for glucose synthesis or ketone body formation. Sources and fates of amino acids are shown in Fig. 12.1.

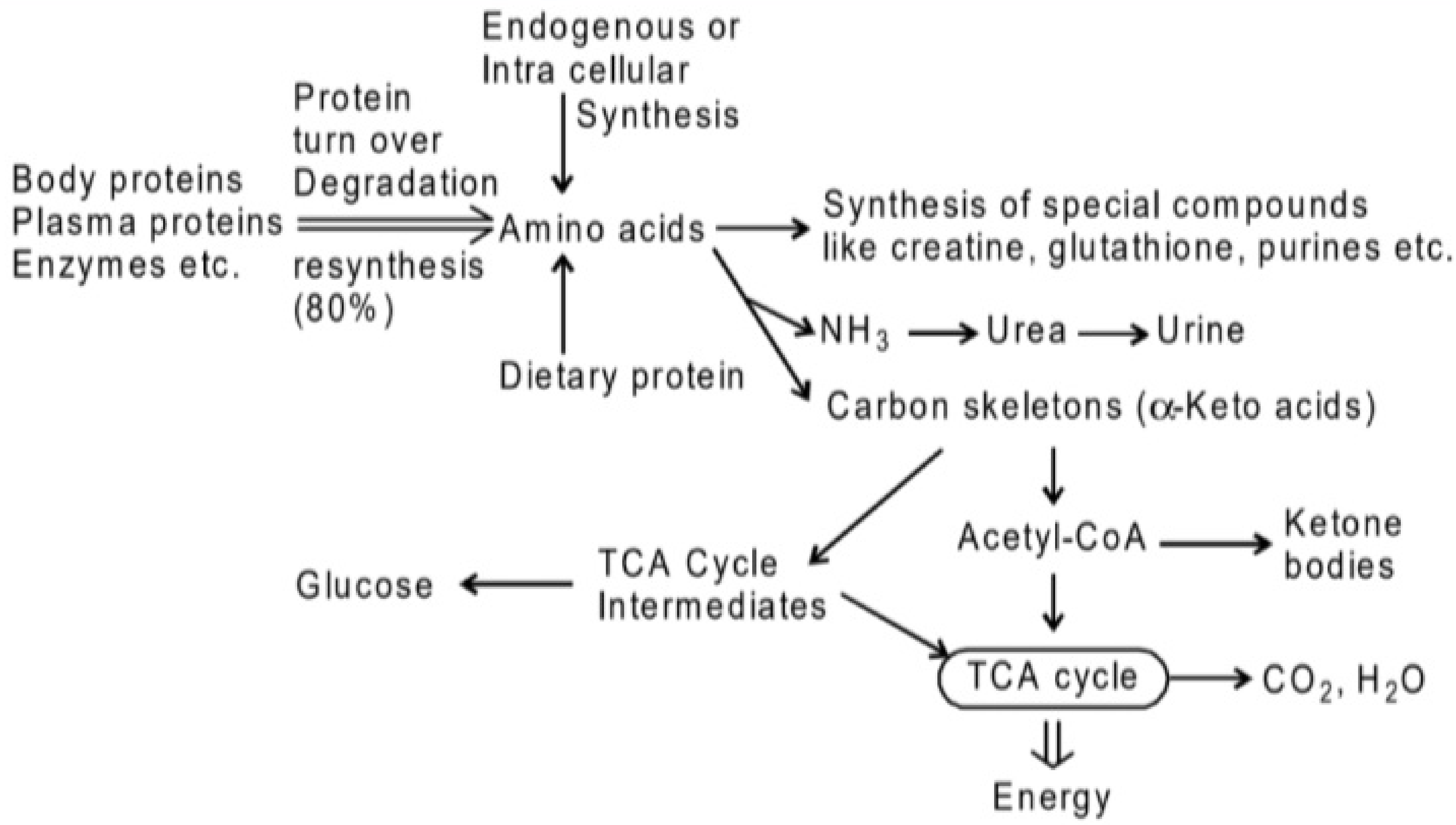


Fig. 12.1 Sources and metabolic fates of amino acids

DEAMINATION OF AMINO ACIDS:

Since the removal of α-amino group is the first stage of amino acid degradation we shall see now how it occurs. There are several ways for the removal of α-amino group of amino acids. They are:

(1) Transamination followed by oxidative deamination

(2) Oxidative deamination

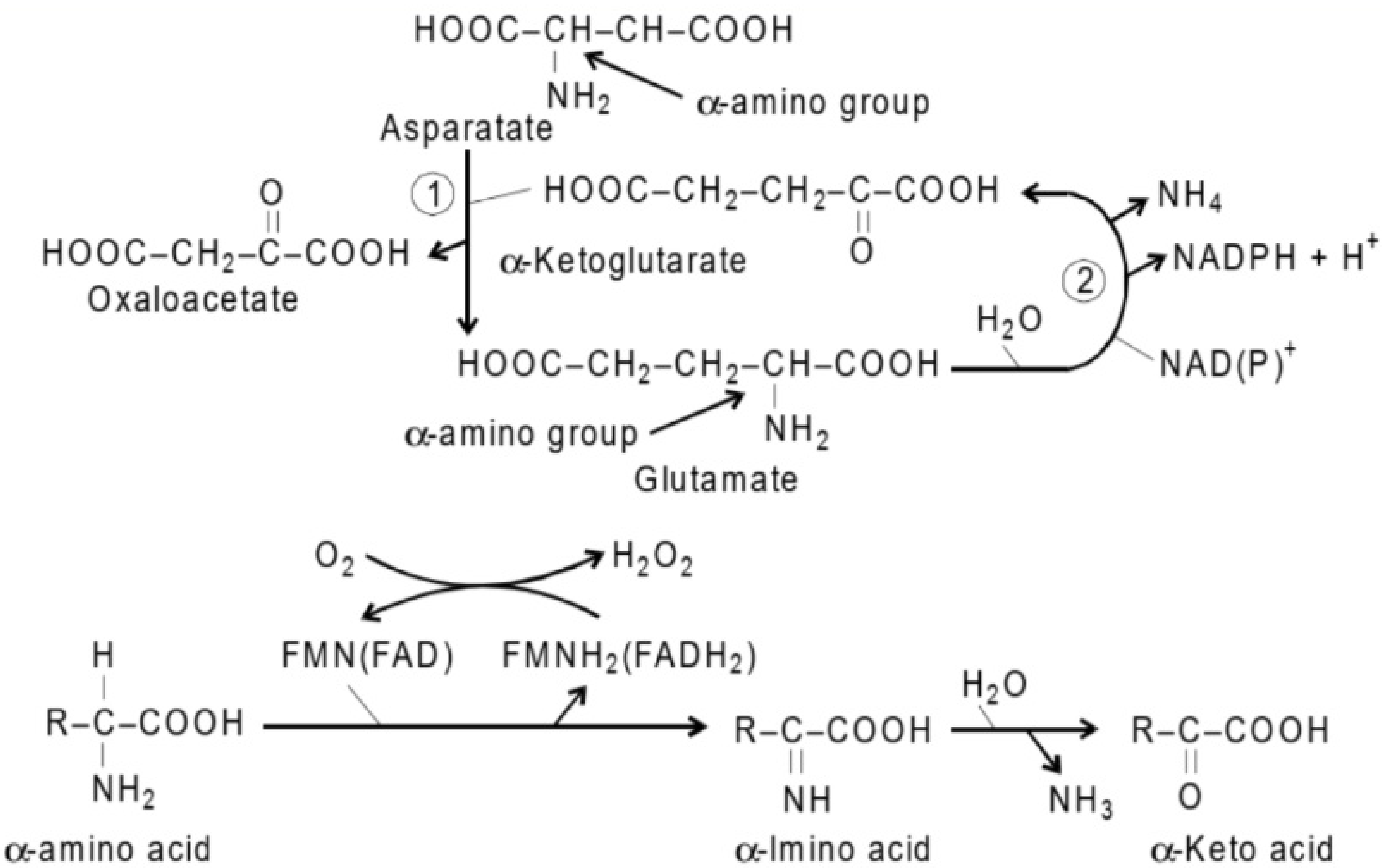
(3) Non-oxidative deamination.

TRANSAMINATION FOLLOWED BY OXIDATIVE DEAMINATION:

a) TRANSAMINATION:

Removal of amino group of amino acids by transamination is the first step in the catabolism of most of the aminoacids. The enzymes involved in this process are known as transaminases. They transfer α-amino group to an acceptor mostly to α-keto glutarate. They are named according to substrate whose aminogroup they transfer to α-keto glutarate. For example, (Fig. 12.2a) aspartate transaminase transfer (removes) amino group of asparatate to α-ketoglutarate. This results in formation of oxaloacetate which is α-keto acid of aspartate and glutamate, which is the corresponding amino acid of α-keto glutarate (a keto acid).

Transaminases are present in mitochondria and cytosol of most of the tissues. They contain pyridoxal phosphate as prosthetic group. Though there are more than ten transaminases most important among them are aspartate amino transferase (transaminase) and alanine amino transferase. So, by the action of transaminases α-amino groups of most of the amino acids are transferred to α-ketoglutarate to form glutamate. Thus glutamate serves as a collecting point of α-amino groups.



Fig, 12.2 (*a*) Reactions catalyzed by (1) Aspartate amino transferase (2) Glutamate dehydrogenase
(*b*) Reaction catalyzed by amino acid oxidase

b) OXIDATIVE DEAMINATION:

Amino groups that are collected by glutamate are removed as ammonia by enzyme glutamate dehydrogenase in presence of NAD or NADP+. Glutamate dehydrogenase catalyzes oxidative deamination of glutamate to yield α-ketoglutarate and ammonia (Fig. 12.2a). The enzyme is present in mitochondria and cytosol of liver. It is an allosteric enzyme. ATP and GTP are allosteric inhibitors whereas ADP and GDP are allosteric activators.

Thus, the combined action of transaminases and glutamate dehydrogenase results in the net removal of amino groups of most amino acids as ammonia.

OXIDATIVE DEAMINATION:

Amino acids undergo NAD+ independent oxidative deamination also. Amino acid oxidases are the enzymes which catalyzes this type of oxidative deamination. They are of two types and dependent on FMN or FAD. They are D-amino acid oxidase and L-amino acid oxidase. Since the activity of L-amino acid oxidase is low its contribution to ammonia production is less.

D-AMINO ACID OXIDASE:

It is present in liver and kidney. It cannot deaminate many amino acids. It uses FAD as cofactor. It can deaminate glycine.

L-AMINO ACID OXIDASE:

It is also present in liver and kidney. It cannot deaminate glycine and dicarboxylic acids. It is dependent on FMN.

The reaction mechanism of these oxidases involves first oxidation of aminoacid to iminoacid. FMN (FAD) is reduced. Next imino acid undergo hydrolytic loss of ammonia to produce α-keto acid (Fig. 12.2b).

NON-OXIDATIVE DEAMINATION:

Non-oxidative deamination of amino acids is catalyzed by specific enzymes. Some of them are given below (Fig. 12.3).

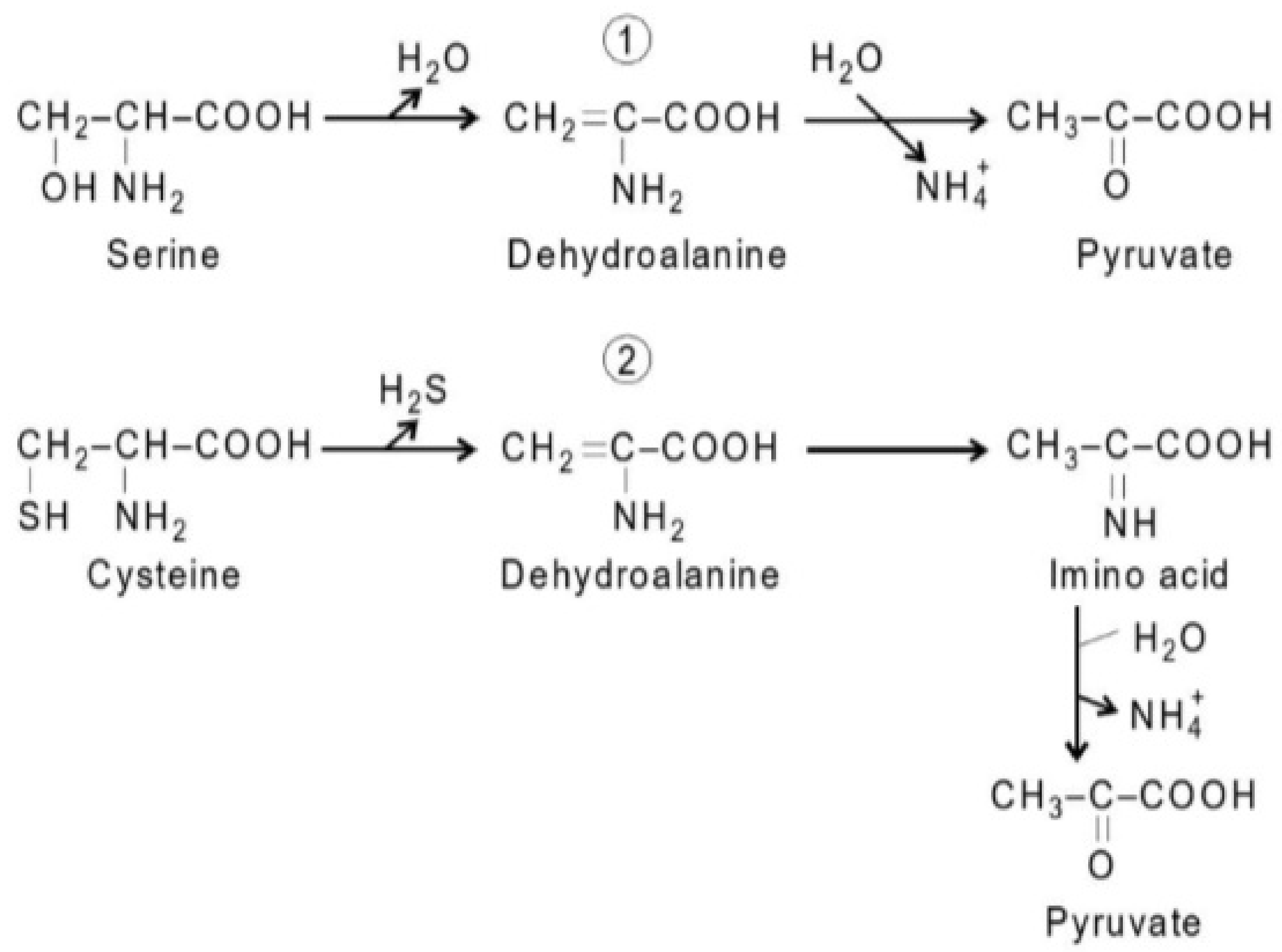
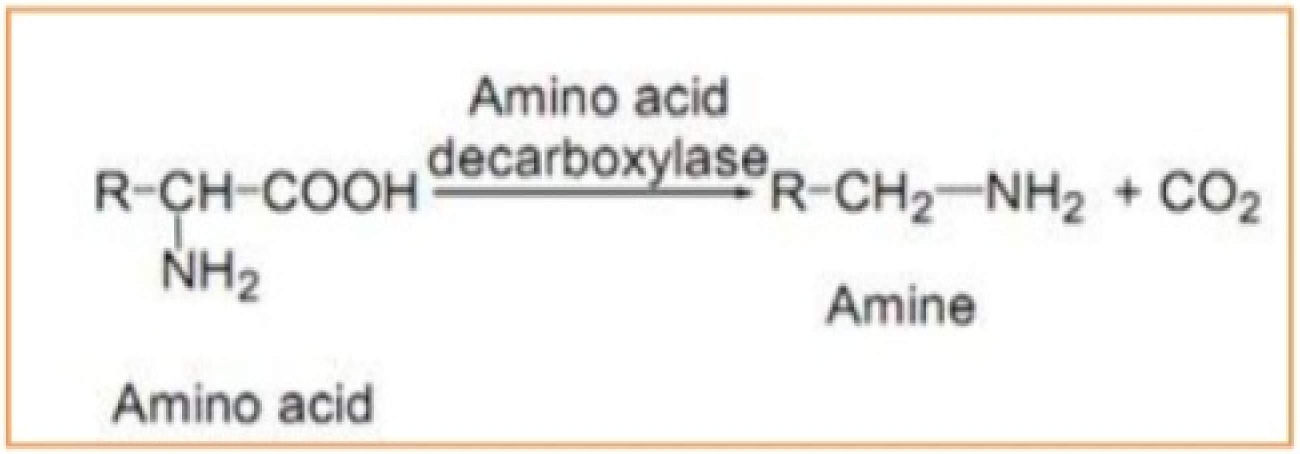


Fig. 12.3 Reactions catalyzed by (1) Serine dehydratase and (2) Cysteine de Sulfhydrase

1. Serine dehydratase It catalyzes nonoxidative deamination of serine to pyruvate. The reaction mechanism involves dehydration of serine to dehydro alanine followed by hydrolytic loss of ammonia. It requires pyridoxal phosphate as cofactor.
2. Cysteine desulfhydrase It is present in bacteria and requires pyridoxal phosphate as cofactor. It catalyzes conversion of cysteine to pyruvate. No such desulfhydrase is found in animal tissue. The reaction mechanism involves desulfuration of cysteine to dehydroalanine followed by hydrolytic loss of ammonia from aminoacid. Dehydroalanine tautomerizes to iminoacid.
3. Threonine dehydratase catalyzes non-oxidative deamination of threonine All the above enzymes are involved in the removal of α-amino group nitrogen as ammonia. Now we shall see removal of amide nitrogen of glutamine and aspargine.

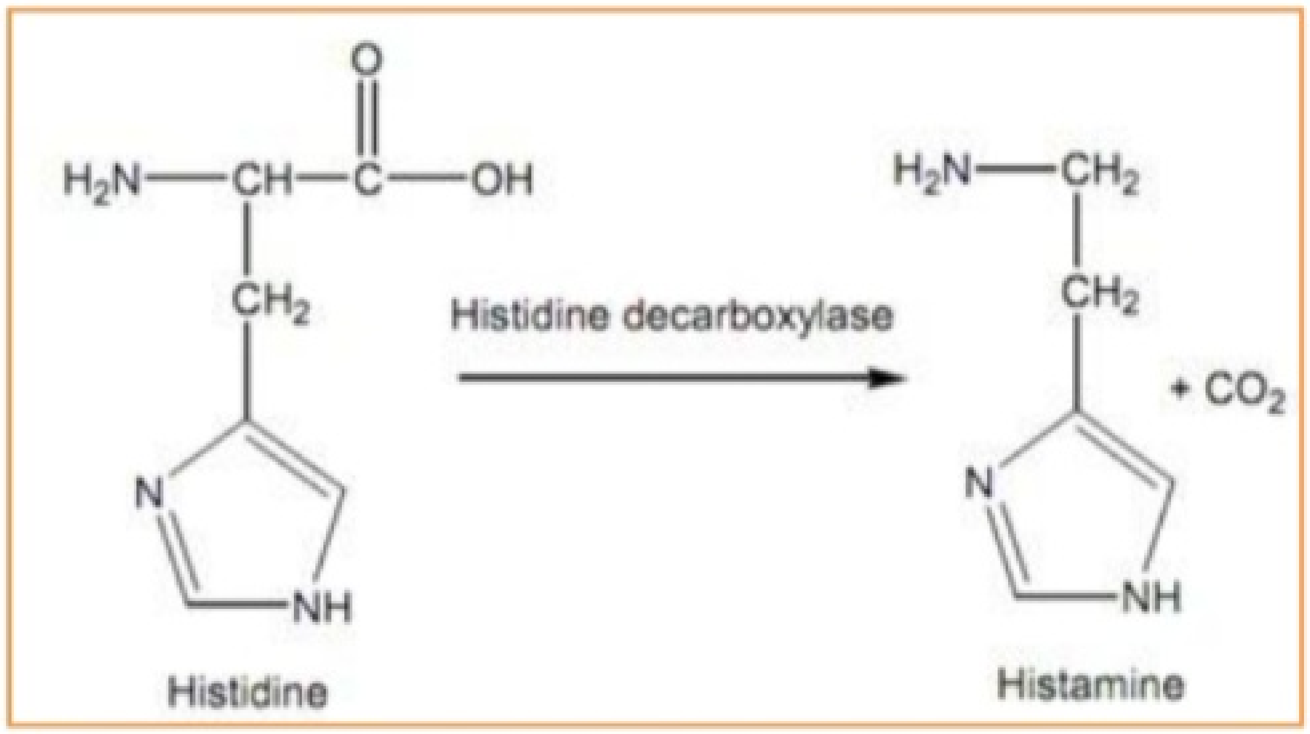
DECARBOXYLATION:

This consider to the removal of CO_2 (carbon dioxide) from the carboxyl group of amino acids. The elimination of CO_2 requires the catalytic action of the pyridoxal phosphate coenzyme and enzymes decarboxylases. The enzymes work on amino acids resultant in the creation of the corresponding amines along with the liberation of CO_2.



There are various amino acid decarboxylases make out in several tissues like liver, spleen, kidney, intestine, lung and brain. They translate the amino acids into the relevant amines and liberate CO_2. For instance, histidine is transformed to histamine via the action of histidine decarboxylase.

The amino acid tryptophan is transformed to tryptamine, tyrosine to tyramine, etc. These types of amines are known biogenic amines that are physiologically significant.


H2N—CH—C—OH
CH2
Histidine decarboxylase
H2N—CH2
CH2
+ CO2
N
NH
Histidine
Histamine